

7. Analytics Dashboard Module

- Disease prediction analytics
 - Symptom trend analysis
 - Healthcare insights
 - System performance monitoring
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5. Week-wise Module Implementation and High-level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

- Define healthcare workflows and project objectives.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Load Disease Symptoms and Patient Profile Dataset, Disease Prediction Using Symptoms Dataset, and CDC BRFSS Dataset for symptom analysis, disease prediction, and risk assessment workflows.
- Build symptom collection and patient management workflows.

Recommended Healthcare Datasets

1. Disease Symptoms and Patient Profile Dataset (Kaggle)

Purpose:

- Symptom-to-disease prediction
- Disease classification
- Medical recommendation workflows

Contains:

- Disease labels
- Symptom indicators
- Patient condition mapping

Example Diseases:

- Diabetes
- Malaria
- Dengue
- Hypertension
- Asthma

Usage:

- Disease Prediction Module
- Symptom Analysis Module

2. Disease Prediction Using Symptoms Dataset

Purpose:

- Multi-class disease prediction
- Symptom correlation analysis

Contains:

- 100+ symptoms
- Disease categories
- Symptom combinations

Usage:

- Disease Prediction Module
- Risk Assessment Module

3. CDC Behavioral Risk Factor Surveillance System (BRFSS)

Purpose:

- Health risk assessment
- Population health analytics

Contains:

- Lifestyle indicators
- Health conditions
- Demographic information

Usage:

- Risk Assessment Module
- Healthcare Analytics Dashboard

4. MIMIC-IV Dataset (Optional Advanced Dataset)

Purpose:

- Clinical decision support
- Healthcare analytics
- Patient risk prediction

Contains:

- Electronic health records
- Diagnoses
- Patient observations
- Clinical measurements

Usage:

- Advanced healthcare analytics
- Research-oriented implementations

Note:

This dataset requires credentialed access and may not be necessary for student-level implementations.

Dataset-to-Module Mapping

Dataset	Symptom Analysis	Disease Prediction	Risk Assessment	Analytics Dashboard
Disease Symptoms Dataset	Yes	Yes	Yes	No
Disease Prediction Dataset	Yes	Yes	Yes	No
CDC BRFSS	No	No	Yes	Yes
MIMIC-IV	Yes	Yes	Yes	Yes

Primary Dataset Recommendation

For project implementation, the recommended dataset is:

Disease Symptoms and Patient Profile Dataset (Kaggle)

Reason:

- Easy to access
- Clean symptom-to-disease mapping
- Suitable for classification models
- Appropriate for academic and student projects

Outcomes

- Understand AI applications in healthcare and disease prediction.
- Learn system architecture and database design concepts.
- Build frontend and backend project initialization.
- Working authentication and symptom management system.

Milestone 2: Week 3 & 4 — Disease Prediction & Risk Assessment

- Train disease prediction models.
- Evaluate model performance and accuracy.